

Kevin S. Niu

Atlanta, GA | 470-636-5696 | niukevins@gmail.com | U.S Citizen

Education

Georgia Institute of Technology | Atlanta, GA

Bachelor of Science in Computer Engineering

May 2024 – Present

Expected Graduation, May 2027

University of Illinois at Urbana-Champaign | Champaign, IL

Transfer with 53 Credit Hours

August 2023 – May 2024

Experience

RoboJackets | Atlanta, GA

May 2024 – Present

Combat Robotics Finance Chair, 12lb Chassis Lead, 3lb Team Captain

- Managed \$9k budget for 100+ members across 9 teams, securing and allocating funding from corporate sponsors and the Student Organization Finance Office while managing reimbursements and purchase requests through Ramp.
- Oversaw procurement for all competition teams, meeting with Georgia Tech accountants weekly to coordinate purchasing and delivery of sensitive electronics and raw materials to off-campus workshop.
- Led interdisciplinary team of ~10 members to design and manufacture a 12lb combat robot while teaching CAD, soldering, and manufacturing fundamentals to less experienced members.
- Won 3rd out of 21 robots at local tournament and competed nationally as a rookie 3lb robot with a budget of only \$500.
- Designed a hub-motor drum spinner system using an integrated sensor-less brushless powertrain, achieving a 12% weight reduction and 15% higher energy storage vs. previous belt-driven designs.

Illinois Electric Motorsports | Champaign, IL

August 2023 – May 2024

Circuit Design Team

- Designed a 4-layer vehicle sensor interface PCB in Altium, using STM32 MCU and CAN protocol for distributed telemetry.
- Implemented RC low-pass filters and operational amplifiers for signal conditioning of analog sensors.
- Led 3-member validation sub-team developing an Arduino-based current-RPM test rig using a Hall-effect current sensor, achieving 20% current draw reduction through fan curve optimization.
- Oversaw assembly of high-voltage accumulator modules using COSMX pouch cells, thermistor harnesses, and voltage sense boards; validated with oscilloscope, logic analyzer and CAN telemetry logs to confirm zero packet loss.

iRobotics | Champaign, IL

August 2023 – May 2024

Combat Robotics Electronics Lead, RoboBrawl Committee

- Developed embedded wireless system with ESP32 and ESP-NOW to display wins/losses and ready status over asynchronous webserver using AJAX for judge-competitor communication, reducing match downtime by 20%.
- Directed a team of 10+ members to source and integrate electronics for a 30-pound combat robot.
- Tuned firmware parameters to enhance drivetrain consistency, reducing motor failure rate and improving startup commutation.

Projects

OpenTurner

July 2025

Embedded System Design (ECE4180)

- Built a battery-powered ESP32-C3 wireless page-turner with dual foot pedals, 5-way nav switch, and a SSD1306 OLED UI, using C++ with non-blocking event handling, hardware debouncing, and persistent NVS configuration.
- Added ADC-based battery monitoring, sleep-mode power management with GPIO wake, and dynamic device naming.

Target Tracking Turret

January 2025

Personal

- Developed a real-time tracking turret firing Nerf HIR rounds at >100 FPS, using OpenCV for HSV color tracking and centroid-based aim correction with closed-loop PID control for servo actuation and flywheel speed regulation.

Analog PID Car

May 2024

UIUC James Scholar Honors Lab (ECE198)

- Designed analog PID control circuitry using op-amps for proportional, integral, and derivative feedback for autonomous braking and distance control, tuning constants via oscilloscope analysis to achieve <2% steady-state error and zero overshoot.

Skills

Programming: Java, Python, C/C++, JavaScript, Node.js, SQL

Developer Tools: Git, Docker, VS Code, PyCharm, IntelliJ, Eclipse

Embedded Systems: STM32, ESP32, Arduino, Raspberry Pi, FreeRTOS, CAN bus, I²C, SPI, UART, PWM

Hardware: VHDL, Verilog, FPGA, Altium Designer, Altera Quartus

Mechanical: SOLIDWORKS (CSWA), Fusion 360 CAM, Inventor, 3D Printing, 2-Axis Mill, Lathe, CNC Router, Waterjet

Hobbies: Pickleball, Poker, Photography, Skating, Woodworking